SRI GANESH COLLEGE OF ENGINEERING & TECHNOLOGY

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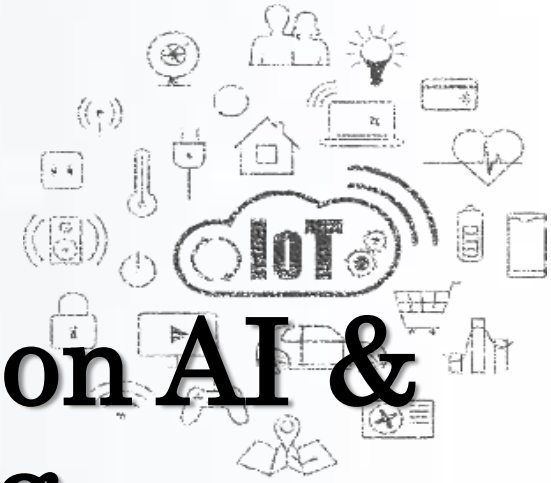
Gladly Presents,



Free Summer Course on AI & Ethical Hacking

DAY 4

CYBER SECURITY

&

INTERNET OF THINGS

CYBER SECURITY

WHAT IS CYBER SECURITY ?

- Cyber Security is the practice of protecting systems, networks, and programs from digital attacks.
- These cyber attacks usually aim to access, change, or destroy sensitive information, extort money, or interrupt normal processes.

IMPORTANCE OF CYBER SECURITY

- Protects Personal Information
- Safeguards Businesses
- Prevents Financial Loss
- Maintains Privacy
- Supports National Security
- Ensures Safe Online Transactions
- Protects Against Cyber Threats
- Enhances Customer Trust
- Supports the IT Team
- Assists the Remote Workspace
- Regulatory Compliance

FUNDAMENTAL MODEL

Confidentiality: **Keeping data private and accessible only to authorized users.**

Example: Passwords protect your email.

Integrity: **Making sure data is accurate and not altered without permission.**

Example: A bank statement shows the correct balance.

Availability: **Ensuring systems and data are available whenever needed.**

Example: A hospital must access patient records 24/7.

COMMON CYBER THREATS

- Malware – Viruses, worms, ransomware
- Phishing – Fake emails/websites to steal information
- Ransomware – Locks files, demands money
- Man-in-the-middle – Intercepts communication
- Social Engineering – Tricking people into giving access

TYPES OF HACKERS

- White Hat – Ethical hackers who help protect systems
- Black Hat – Malicious hackers with harmful intent
- Grey Hat – Hackers who might break laws but not maliciously
- Script Kiddies – Amateurs using tools without real understanding

CYBER SECURITY CAREER PATHS

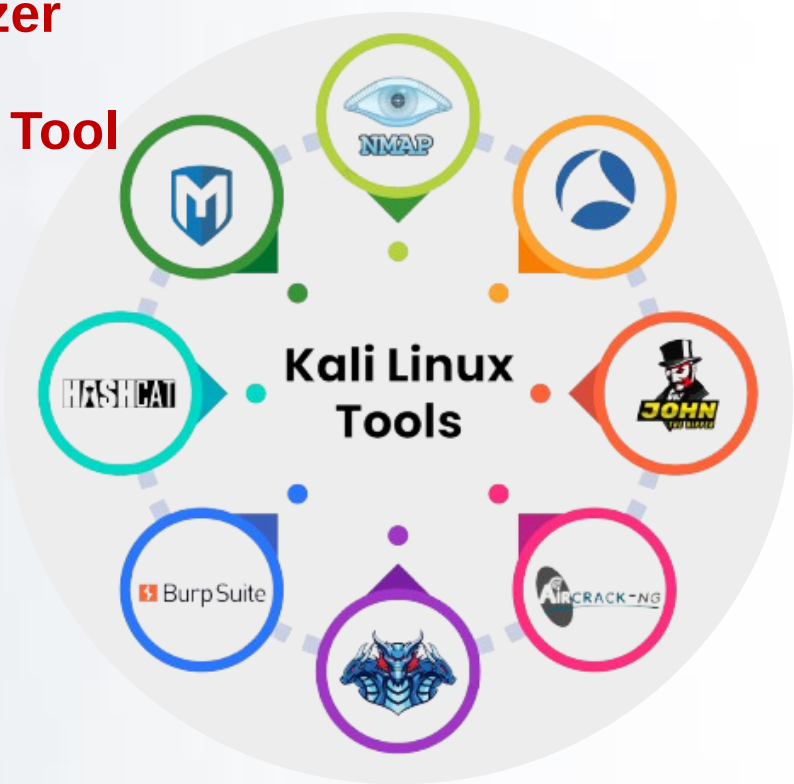
- Security Analyst
- Penetration Tester (Ethical Hacker)
- Network Security Engin Peer
- Cyber Forensics Expert
- Incident Responder

KALI LINUX

- Kali Linux is a free, open-source Linux-based operating system used mainly for penetration testing, ethical hacking, and digital forensics.
- Developed and maintained by Offensive Security.
- Comes with 600+ pre-installed tools for hacking and security testing.
- Used by ethical hackers, cybersecurity students, and professionals.

KALI LINUX TOOLS

- Nmap – Network Scanner
- Wireshark – Network Protocol Analyzer
- Metasploit Framework – Exploitation Tool
- Air crack-ng – Wi-Fi Cracking Tool
- Hydra – Password Cracking Tool
- John the Ripper – Password Cracker



INTERNET OF THINGS

WHAT IS IoT?

- IoT stands for Internet of Things.
- It refers to interconnected devices that can collect and exchange data over the Internet.
- These devices can sense, communicate, and act.

Examples:

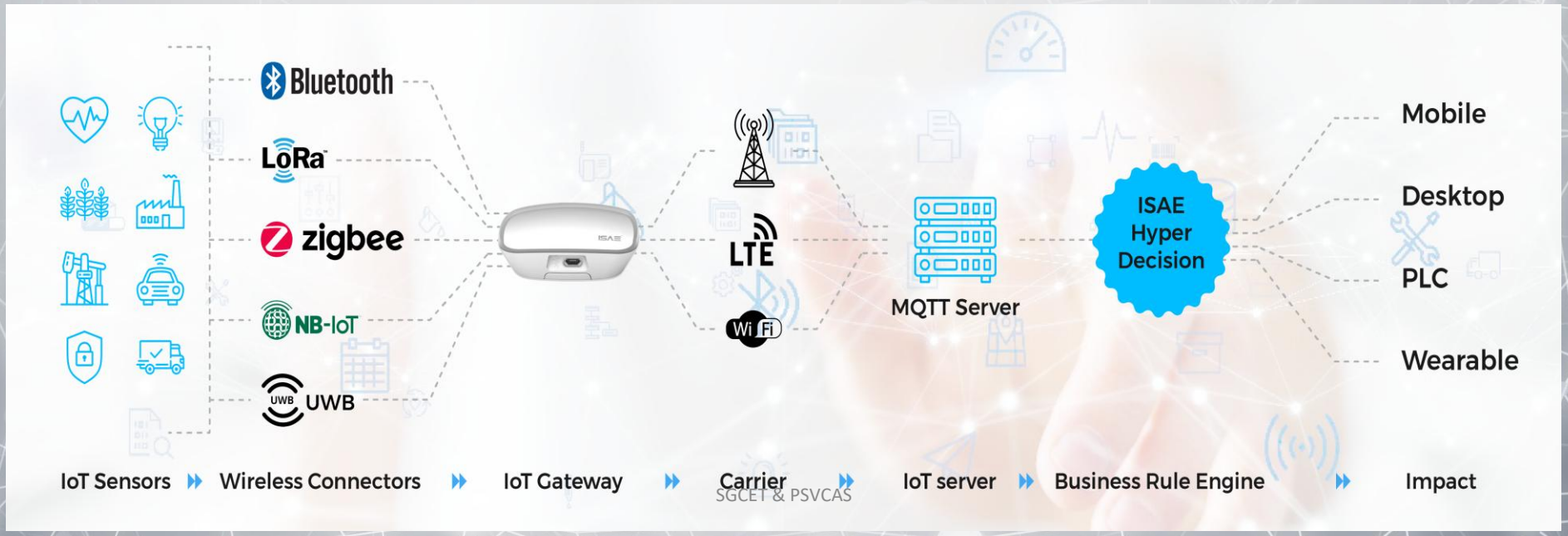
- Smart Bulbs
- Fitness Bands
- Smart Thermostats
- Voice Assistants (Alexa, Google Home)

WHY IoT MATTERS ?

- Automates daily tasks
- Improves efficiency and convenience
- Helps make data-driven decisions
- Key in smart homes, smart cities, healthcare, agriculture, etc.

HOW IoT WORKS ?

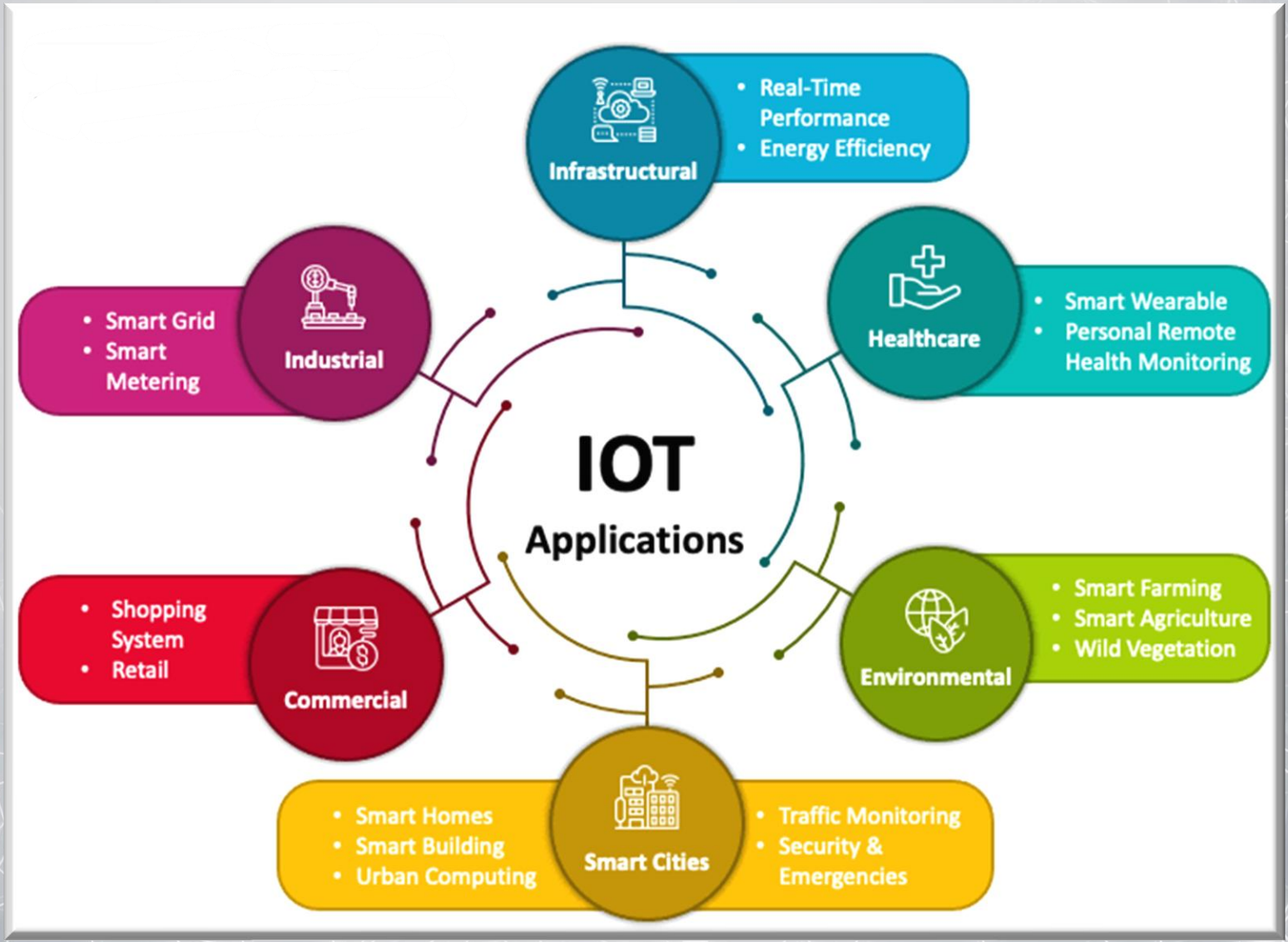
- IoT Sensors collect data from the environment.
- Data is sent wirelessly using Bluetooth, LoRa, Zigbee, NB-IoT, etc.
- IoT Gateway receives the data and passes it to the network.
- Through Wi-Fi or LTE, the data travels to the IoT Server.
- The Business Rule Engine processes the data and makes smart decisions.
- Final results or actions are sent to Mobile, Desktop, Wearable, or PLC devices.



COMPONENTS OF AN IoT SYSTEM

- Sensors/Devices – Collect real-world data.
- Connectivity – Transfers data via Wi-Fi, Bluetooth.
- Microcontroller/Processor – Interprets and processes data.
- Cloud/Server – Stores and analyzes data.
- User Interface – App/web dashboard to monitor or control

APPLICATIONS OF IoT



THANK
YOU